JIAYI MA 马佳艺

Master's candidate in Physical Education and Training, Beijing Sport University

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RESEARCH INTERESTS

Markerless motion capture and computer vision in sport · expert-aligned large language models for coaching feedback · injury-risk movement patterns in youth sabre fencing · motor skill development and analysis

EDUCATION

Beijing Sport University — Beijing, China — Sep 2024 – Jun 2027 (expected)

Master's programme in Physical Education and Training (research track), School of Competitive Sports

- Thesis: *Computer vision and large language models for injury-risk assessment in youth sabre fencing*. Supervisor: Prof. Liqing Zhang
- Coursework: principles of sports training, advanced statistics, experimental research methods, advanced sport psychology, sports medicine

Springfield College — Springfield, MA, USA — Aug 2025 – Jun 2026

Visiting Scholar, funded by the China Scholarship Council (CSC)

- Big Data in Sport Science; Motor Skill Development and Analysis; Evidence-Based Practice (all A grades). Passed MTEL Physical Education (22).

The Chinese University of Hong Kong — Hong Kong, China — Jun – Jul 2023

Short-term academic exchange, Sports Training — volleyball, karate, cycling (all A grades)

Beijing Sport University — Beijing, China — Sep 2020 – Jun 2024

Bachelor's degree in Sports Training

- Ranked 2 of 186 in cohort; Beijing Municipal Outstanding Graduate (2024). Coursework included sports medicine, sports training theory, sport psychology, and PE curriculum and pedagogy.

RESEARCH EXPERIENCE

Master's thesis — Beijing Sport University — 2024 – present

- Five-stage pipeline for sabre fencers aged 12–18: YOLOv8-Pose keypoints from dual-camera 2D video → 15 biomechanical indicators (en garde stance, sagittal- and frontal-plane lunge kinematics, inter-segment timing) → supervised fine-tuning (SFT) → preference alignment (DPO) of Qwen2.5-7B with LoRA → interpretable injury-risk reports.
- The language model is aligned to expert coaches' judgements. Ethics approval obtained from Beijing Sport University; evaluation against independent expert-coach ratings is in progress.

Haoli (毫厘) · Proof of Call — independent project — Aug 2026 – present

- Evidence-chain system for sabre right-of-way calls: aligns skeletal time series with broadcast footage frame by frame and links each call to the relevant FIE rule, making calls auditable rather than replacing referees.
- Evaluated on contested calls from the Paris 2024 Olympic Games; being integrated into the Haidian District Fencing Association's official referee-training platform. Selected for the software track of the SheNicest hackathon (1,000 participants). github.com/majiayi0526/haoli-proof-of-call

Deep Court — BroadcastCode 12 hackathon (NBA × AWS × CSDN); team captain, AI runner-up — Oct 2026

- 12-hour build of an AI broadcast system: players are identified with YOLO detection, OSNet re-identification and jersey numbers checked against the official roster; an agent on Amazon Bedrock AgentCore plans graphics and commentary, voiced with Amazon Polly. Unconfirmed identities are never named, and metrics are shown side by side rather than as causes.

Other applied projects

- **Pickleball skill analysis** (PE Department, Beijing Information Science and Technology University, 2026): MediaPipe and dynamic time warping compare strokes against 8 reference templates and return scores, parameter deviations and corrective feedback. pits-v1.vercel.app
- **Expert chain-of-thought simulation for épée training** (Beijing Sport University fencing team): a senior coach's diagnostic pathway as an executable chain of thought. epee-coach-tan.vercel.app
- **Fencing Movement Recognition Mini-Program V1.0**: registered software copyright (China); model training and development.
- **Game-based activities in junior-high PE**: research group member; First Prize, PE Teaching & Research Award.

MANUSCRIPT IN PREPARATION

Ma, J., et al. *Concurrent validity of single-camera open-source pose estimation for joint-angle measurement against marker-based motion capture: a systematic review and meta-analysis.*

SELECTED CONFERENCE PRESENTATIONS *10+ in total, 2024–2026*

1. Ma, J. (2026, April). *Who coaches the coach? Augmenting pedagogical decision-making through expert-aligned AI in adolescent sabre training* [Invited talk]. 17th Harvard China Education Symposium (student-organised), Cambridge, MA, USA.
2. Ma, J., & Zhang, L. (2026, March). *Developmental sequence of sabre lunge technique* [Oral presentation]. ICSPAH 2026 Annual Conference, International Chinese Society for Physical Activities and Health.
3. Ma, J., & Zhang, L. (2025, November). *Markerless motion-capture-based early-warning model for lower-limb injuries in sabre fencers* [Oral presentation, in Chinese]. 14th National Congress on Sports Science, China.
4. Ma, J., & Zhang, L. (2025, June). *Deep-learning-based precision recognition and analysis system for sabre techniques* [Oral presentation, in Chinese]. 2nd National Conference on Artificial Intelligence in Sports, China.
5. Ma, J., & Zhang, L. (2025). *Performance-training games improve fencing techniques by affecting physical and cognitive function in high-school athletes*. Asian Society for Coaching Science Conference (ASCS 2025).
6. Ma, J., Qu, R., Xu, L., … Zhang, L. (2024, December). *Deep-learning-based accurate recognition and analysis of sabre movement* [Oral presentation]. ICSTPA 2024 (EI-indexed proceedings).
7. Ma, J. (2024, April). *Effects of experimental game-based training on promoting physical fitness of high-school fencing athletes* [Poster]. International Symposium on Embodied Cognition, Yale Center for Brain and Mind Health, New Haven, CT, USA.

COACHING AND OFFICIATING

Head Coach, Sabre — The High School Affiliated to Peking University, Beijing Sep 2020 – present

- Head sabre coach for the primary, junior-high and senior-high divisions, about 20 student-athletes per cohort over five academic years; coached athletes to 4 gold and 2 silver medals at national level and 5 gold and 3 silver medals at Beijing municipal level.
- Co-authored the school's first fencing curriculum (junior-high PE elective), adopted as the standard teaching material.

Grade-A Referee — Chinese Fencing Association 2020 – present

- 22 events officiated: 1 national-level, 3 Grade-A, 2 Grade-B and 16 Grade-C/D events. Also a certified tennis referee (Chinese Tennis Association, 2025), including the FILA KIDS Diamond Cup Junior Tennis Challenger (CTJ-B600).

ATHLETIC RECORD

- National Class-I Athlete, fencing (sabre)
- 2nd Youth Games of the PRC, women's sabre individual (Group A) — runner-up, 2019
- 15th Beijing Municipal Games, women's sabre individual (Group A) — champion, 2018
- China Secondary School Fencing Championships, women's sabre individual — runner-up, 2018
- Chinese National Youth Fencing Team training camp — selected, 2018
- Beijing Sport University Fencing Games, women's sabre individual — champion, 2021
- U.S. collegiate women's rugby union — starting player; team placed third, 2025–26
- Peking University "Weiming Bowl" Flag Football Tournament — champion, starting player, 2025

HONOURS AND SCHOLARSHIPS

- National Scholarship (China), Ministry of Education
- China Scholarship Council (CSC) award, visiting scholar at Springfield College, 2025–26
- Henry Scholarship, Springfield College
- Beijing Merit Student, Beijing Municipal Education Commission
- Beijing Municipal Outstanding Graduate, 2024
- AI runner-up, BroadcastCode 12 hackathon (NBA × AWS × CSDN), 2026
- National finalist, 19th Advanced Robotics & Simulation Technology Competition, 2026
- Outstanding Participant, Hunan Provincial Graduate Summer School on AI for National Fitness & Health, 2026

CERTIFICATIONS AND SKILLS

Officiating	Grade-A Referee, Chinese Fencing Association; USA Fencing Referee (P rating, sabre); USA Boxing Certified Official (Level 1); tennis, track and field, and sanda referee (China)
Teaching	Passed MTEL Physical Education (22), Massachusetts; High School PE Teacher Qualification (China); NSCA Tactical Strength and Conditioning Facilitator (TSAC-F); First Aid / CPR / AED
Vision & ML	Python, PyTorch, OpenCV, YOLOv8-Pose, MediaPipe; LLM fine-tuning with LoRA / LlamaFactory (SFT, DPO)
Methods	R, SPSS, SQL; reliability and agreement analysis (ICC, weighted kappa); expert-panel design; markerless motion capture
Prototyping	Full-stack builds with AI coding assistants: React + TypeScript, Streamlit, AWS, Vercel
Languages	Mandarin Chinese (native); English (CET-6)
Sports taught	Fencing, pickleball, flag football, boxing, badminton, swimming

SERVICE

Outstanding Volunteer, SHAPE America National Convention & Expo (2026); volunteer at the World Para Fencing Games and the ICSPAH Annual Conference (2026), the China Open tennis tournament (2023; Gold Award for volunteer service, 110 hours) and the China International Fair for Trade in Services (2023).